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A Graphical Tool to Examine the Coordination Details of Directional Overcurrent Protections as a Function of Fault Location

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Abstract—This letter describes a novel graphical tool that was developed to check the coordination details of directional overcurrent protections (67) as a function of the fault location. The tool considers transient configurations due to sequential trips at both line-ends, and shows three important issues of coordination for each fault location: a) separation between operation times of main and backup 67s; b) currents seen by each 67 prior to and after the trip of the first circuit breaker; c) possible loss-of-sensitivity of backup 67s due to the pickup setting or due to direction of current. A previously developed simulator performs the calculations, using 1000 fault locations for each protected line, and it is complemented by the proposed graphical tool.

Index Terms—Directional overcurrent protection.

I. INTRODUCTION

THE coordination of directional overcurrent protections (ANSI code: 67) has been analyzed since much before the first application of optimization tools for that purpose [1], but the publication of papers about this topic has been notoriously greater after that. Coordination of 67s has different interesting details [2], such as the effect of transient configurations due to sequential trips at both line-ends [3], [4] and the effect of fault location [5]. Although these effects have been previously analyzed, many papers about this topic do not consider them and, consequently, cannot offer a complete perspective of the coordination. An assessment tool was developed [6], considering both issues, and a detailed classification of the possible behavior of main-backup 67s' pairs was shown. For example, the backup 67 can be insensitive due to the pickup setting (I_{PU}) or due to the reverse direction of the current (I_{REV}), or the time interval between main and backup 67s can be below or above a given safety margin depending on fault location. This letter presents an efficient graphical tool, that was developed to examine the 67s' coordination details as a function of fault location, considering transient configurations due to sequential trips at both line-ends.

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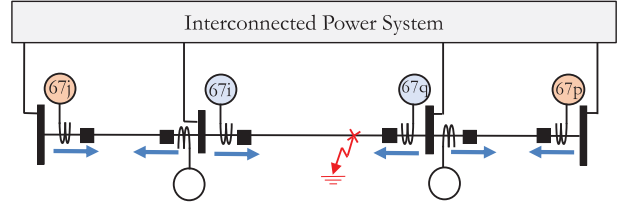


Fig. 1. Sketch to illustrate the nomenclature. The 67 that trips first is 67q.

There is no precedent, in the literature, about graphical tools with similar features.

II. COORDINATION DETAILS TO BE SHOWN

A. Currents Prior and After the Trip of the First Circuit Breaker

Fig. 1 shows an example of a short-circuit in a transmission line of a meshed power system. 67j is a backup of the main 67i, and 67p is a backup of the main 67q. The main 67 at each line-end (67i, 67q) can have more than one 67 as backup. In general, the currents in backup and main 67s are different from each other. Furthermore, the currents change due to the circuit breaker that first trips. Herein, the 67 that trips first is 67q, the time interval before the first trip is period 1, and after that is period 2.

B. Classification of DOCP Pairs

A convenient way to summarize the 6 types and 15 cases of 67s' pairs, previously defined in [6], is shown in Table I. Types 3, 5 and 6 are subdivided into cases, according to the specific reasons for the non-advance of 67s. Only type 6 implies that the separation between the operation times of the main and backup 67s cannot be computed.

C. Separation Between Operation Times (Δt)

The separation between operation times of main and backup 67s (Δt) can be computed only for cases 1 to 9 (types 1 to 5). Case 1 only depends on currents during period 1, and the remaining cases depend on currents during both periods.

III. EXAMPLES OR RESULTS

Different optimal solutions have been published for the 67s' coordination of the system shown in Fig. 2. Two solutions from [4], using IEC Very Inverse and Extremely Inverse curves

TABLE I
CLASSIFICATION OF 67s' PAIRS (A DETAILED EXPLANATION OF THIS CLASSIFICATION IS SHOWN IN [6])

Type	Main description	Case	Reason
1	Both 67s of pair q-p advance towards trip during period 1	1	-
2	Both 67s of pair i-j advance towards trip during periods 1 and 2	2	-
3	67j does not advance during period 1	3	I_{PU}
		4	I_{REV}
		5	I_{PU} and I_{REV}
4	67i does not advance during period 1	6	I_{PU}
5	67i and 67j do not advance during period 1	7	I_{PU} (both)
		8	I_{PU} in 67i; I_{REV} in 67j
		9	I_{PU} in 67i; I_{PU} and I_{REV} in 67j
6	67p does not advance during period 1	10	I_{REV}
	67j does not advance during period 2	11	I_{REV}
	67p does not advance during period 1	12	I_{PU}
	67p does not advance during period 1	13	I_{PU} and I_{REV}
	67j does not advance during period 2	14	I_{PU} and I_{REV}
	67j does not advance during period 2	15	I_{PU}

TABLE II
67s' SETTINGS OF ANALYZED EXAMPLES [4]

	67 ₁	67 ₂	67 ₃	67 ₄	67 ₅	67 ₆	67 ₇	67 ₈	67 ₉	67 ₁₀	67 ₁₁	67 ₁₂	67 ₁₃	67 ₁₄
I_{PU} (A)	120	500	240	120	120	180	120	180	120	120	240	600	120	120
TM_{VI}	0.351	0.383	0.536	0.612	0.157	0.634	0.723	0.635	0.157	0.613	0.541	0.313	0.351	0.724
TM_{EI}	0.496	0.478	1.136	1.866	0.136	1.641	1.314	1.656	0.127	1.856	1.052	0.293	0.529	1.221

TM_{VI} and TM_{EI} are the time dial settings for VI and EI curves, respectively.

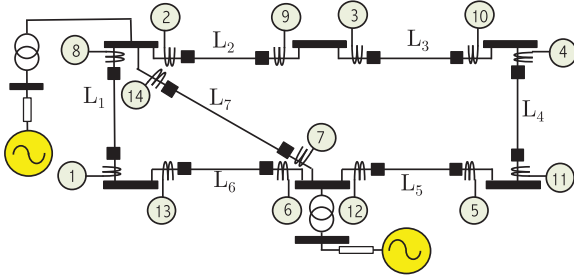


Fig. 2. Power system taken as an example (data are in [7]).

(VI and EI, respectively), were selected for the examples (67s' settings are in Table II); they were found using 0.3s as the selectivity margin S_M) without considering all fault locations. The pickup settings have the maximum possible sensitivity [7].

Fig. 3 shows examples of results with EI curves. In these graphs, d is the distance from the correspondent main 67 up to the fault, in per-unit of the line length, and I_{BU-1} and I_{BU-2} are the currents seen by the corresponding backup 67 during periods 1 and 2, respectively.

Fig. 3(a) shows examples of results for faults in line L_3 . The main 67₃ trips first approximately for d between 0 and 0.42 pu from 67₃ location, and the main 67₁₀ trips first approximately for d between 0 and 0.58 pu from 67₁₀ location. According to the classification of 67s' pairs, only types 1 and 2, and classes 1 and 2 occur. All the contributions of currents seen by the shown backup 67s are decreasing, as a function of d . The separation time Δt is smaller than S_M for the fault locations where the green line is below the dashed line.

Fig. 3(b) shows examples of results for faults in line L_7 . The main 67₇ trips first approximately for d between 0 and 0.52 pu from 67₇ location, and the main 67₁₄ trips first for d between 0

and 0.48 pu from 67₁₄ location. According to the classification of 67s' pairs, types 1, 6 and 3, as well as classes 1, 3, 4, 5, 12 and 13 occur. I_{BU-2} is decreasing as a function of d ; on the other hand, I_{BU-1} first decreases and later increases as a function of d , as a natural consequence of the meshed nature of the network. Δt tends to infinite at $d \approx 0.35$ pu for pair 7-5, and at $d \approx 0.35$ pu for pair 14-9; Δt cannot be computed for d between 0.35 and 0.52 pu, and between 0.35 and 0.48 pu, for the pairs 7-5 and 14-9, respectively.

Fig. 4 shows the corresponding examples of results with VI curves. In comparison with the results of Fig. 3, the values of d for the changes that depend on the curve types are different from each other, whereas the values of d for the changes that only depend on short-circuit currents and 67s' pickup currents are identical in both graphs.

IV. POSSIBLE IMPROVEMENTS

The shown graphs are based on outputs from the simulator described in [6], which correspond to three-phase solid faults, time-delay 67s, and a particular way of directional comparison. Improvements related to the inclusion of different fault types, fault resistances, and other ways of directional comparison can be included in the simulator, and the developed graphical tool simply will show the results. Different pre-fault configurations can be analyzed by running those cases independently.

Inclusion of other protective functions depends on the type of function. For instance, inclusion of instantaneous 67s should be easy because they can trip for some near faults and are a particular case of the 67 that firstly trips (67q). Inclusion of instantaneous line differential protections also seems to be easy since both line-ends can be considered as the circuit breaker that firstly trips, and the backup 67s only can be type 1 or type 6. However, line differential protections can be insensitive for some

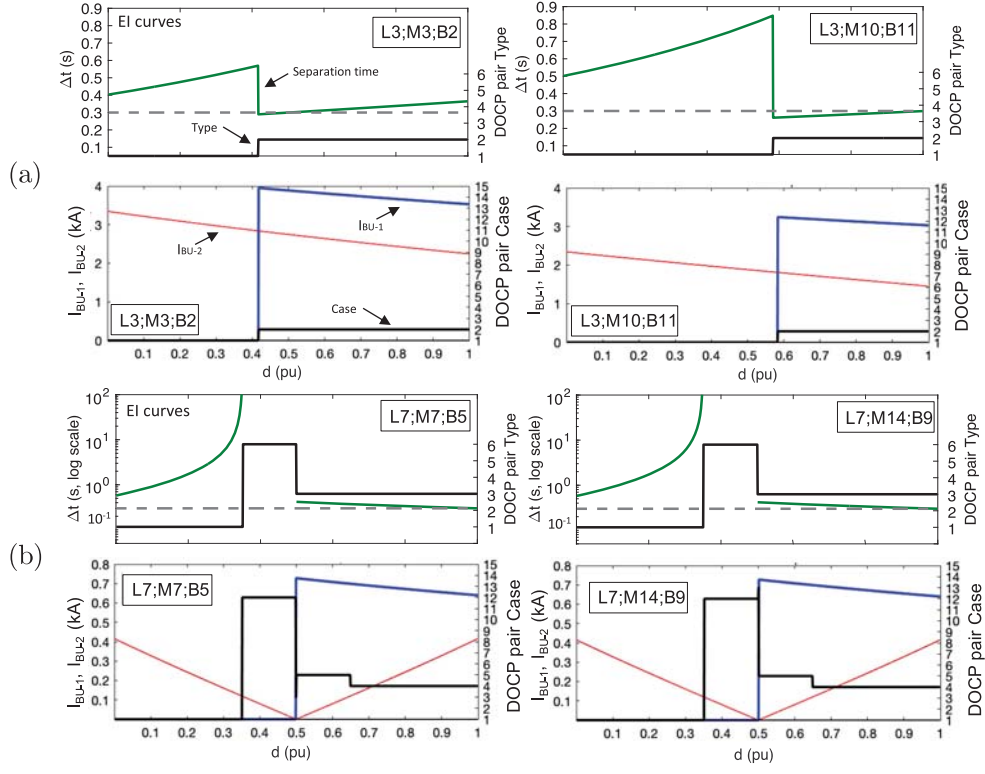


Fig. 3. Examples of results, EI curves. L: line; M: Main; B: Backup; black: case/type; green: separation time Δt ; blue: I_{BU-1} ; red: I_{BU-2} ; dashed: S_M .

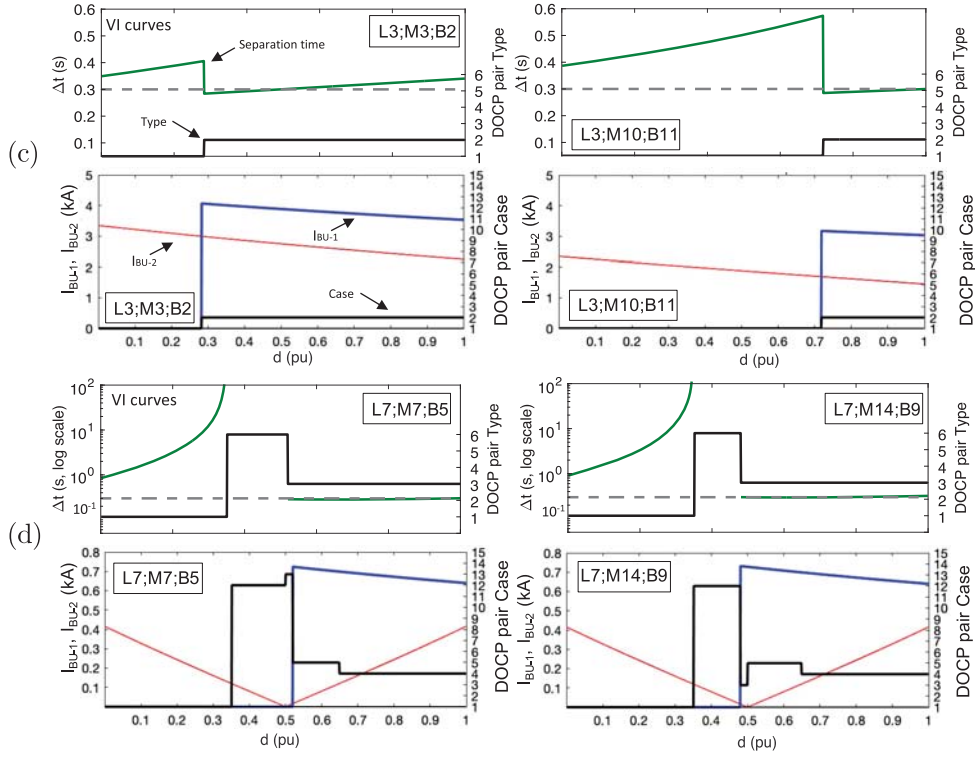


Fig. 4. Examples of results, VI curve. L: line; M: Main; B: Backup; black: case/type; green: separation time Δt ; blue: I_{BU-1} ; red: I_{BU-2} ; dashed: S_M .

high-impedance faults and their behavior usually is dependent on particular details of each relay model. Similarly, inclusion of distance protections does not seem to be evident, especially because the apparent impedances seen by them depend on different factors for each relay model.

In the future, this graphical tool could be complemented with artificial intelligence techniques to offer an advanced help for detecting and solving miscoordination problems.

V. CONCLUSION

This letter presents a novel graphical tool to check the coordination details of directional overcurrent protections as a function of the fault location. The visualization tool considers the transient configurations due to sequential trips at both line-ends, showing issues of the coordination for each fault location, such as possible loss-of-sensitivity of backup directional overcurrent protection due to the pickup setting or due to the direction of the current.

REFERENCES

- [1] A. Urdaneta, R. Nadira, and L. Pérez, "Optimal coordination of directional overcurrent relays in interconnected power systems," *IEEE Trans. Power Del.*, vol. 3, no. 3, pp. 903–911, Jul. 1988.
- [2] E. Sorrentino and N. Gupta, "Summary of useful concepts about the coordination of directional overcurrent protections," *CSEE J. Power Energy Syst.*, vol. 5, no. 3, pp. 382–390, Sep. 2019.
- [3] A. Urdaneta, L. Pérez, and H. Restrepo, "Optimal coordination of directional overcurrent relays considering dynamic changes in the network topology," *IEEE Trans. Power Del.*, vol. 12, no. 4, pp. 1458–1464, Oct. 1997.
- [4] E. Sorrentino and J. V. Rodríguez, "A novel and simpler way to include transient configurations in optimal coordination of directional overcurrent protections," *Electric Power Syst. Res.*, vol. 180, Mar. 2020, Art. no. 106127.
- [5] E. Sorrentino and J. V. Rodríguez, "Effects of the curve type of overcurrent functions and the location of analyzed faults on the optimal coordination of directional overcurrent protections," *Comput. Elect. Eng.*, vol. 88, Dec. 2020, Art. no. 106864.
- [6] P. M. De Oliveira-De Jesus and E. Sorrentino, "Methodology to assess performance indexes for sensitivity, selectivity, and speed of coordination of directional overcurrent protections," *Electric Power Syst. Res.*, vol. 213, Dec. 2022, Art. no. 108758.
- [7] M. Ezzeddine, R. Kaczmarek, and M. Iftikhar, "Coordination of directional overcurrent relays using a novel method to select their settings," *IET Gener., Transmiss. Distrib.*, vol. 5, no. 7, pp. 743–750, Jul. 2011.